

Polynomials (Part I): A Review of Polynomials

Polynomial: A sum of two or more expressions of the form bx^n , where n is a whole number (i.e. 0, 1, 2, 3, ...), b is a constant (called a coefficient), and x is a variable.

Example: $3x^5 + 8x^3 - x^2 + 4x^{-1}$

Warning: The definition above excludes polynomials with different variables (e.g. $5x^2y^3 + 2xy$). We will learn these types of polynomials later in the course.

Determine if the expressions below are polynomials. If not, circle the term(s) that disqualify each expression.

① $x^6 - 8x^4 + 5^x + 2$

No

② $10x^3 - 4x$

Yes

③ $4x^{-2} + 9x^5 - x^{3/4}$

No

④ $-108x^2 + 5x + 40$

Yes

⑤ $3x^2 + 4\sqrt{x} + 6$

No

⑥ $2 + \frac{4}{5}x^{16} + x$

Yes

⑦ $-3x^3 + 1$

Yes

⑧ $8x^5 + 2x^4 - \frac{5}{x^2} + 8x^2$

No

⑨ $42x + x^0 + \pi$

Yes

⑩ $0.002x^6 - 23^{-x/5}$

No

⑪ $-2 + 90x^4 + x$

Yes

⑫ $\frac{3}{x^4} + 5x - x^{-1}$

No

⑬ $6\sqrt[3]{x} + 10x^{-7} + 26$

No

⑭ $\log_6 x - 4x^2 + 3872$

No

⑮ $47x^{101} + 83x^{100} - 3\frac{1}{9}$

Yes

⑯ $\frac{1}{2}x^2 - 70x + \frac{1}{8}$

Yes

⑰ $\frac{x^3}{7} - x^2 + x + 2,000$

Yes

⑱ $500\pi + \frac{3}{8x^4} + 0.14x^2$

No

⑲ $75x^9 + 15x^7 + 3x^0$

Yes

⑳ $36x^{10} - 7(2^{4x})$

No

Simplify each polynomial, write in descending order of powers, and identify the degree.

21. $8x^7 - 2x^5 + 3x^7 + 6x - 9x^5 + 4$

$11x^7 - 11x^5 + 6x + 4$, $D=7$

22. $5x^3 + 3x^4 + 10 + 2x^2 + 6x - 9 + 8x^4 - 2x + 5x - 4x^2$

$11x^4 - 6x^2 + 9x + 1$, $D=4$

23. $2 - x^3 - 12x^2 + 3x^3 + 6x + 1 - 8x + x^2$

$-x^3 - 8x^2 - 2x + 3$, $D=3$

$$(24) 8x + 3x^4 - 5x^3 + x^2 - 3x + 2x^4 + 1$$

$$5x^4 - 4x^3 + 5x + 1, D=4$$

Add the following polynomials.

$$(25) (5x^3 - 2x^2 + 6x - 8) + (3x^2 + 6x - 4)$$

$$5x^3 - 2x^2 + 6x - 8 + 3x^2 + 6x - 4$$

$$5x^3 + x^2 + 12x - 12$$

$$(26) (x^3 - 2x^2 + 3) + (4x^3 - 5)$$

$$5x^3 - 2x^2 - 2$$

$$(27) (-9x^5 + 2x^3 - 6) + (4x^4 - 3x^3 + 8x^2 - 1)$$

$$-9x^5 + 4x^4 - x^3 + 8x^2 - 7$$

$$(28) (5x + 7) + (2x^3 + 6x + 7)$$

$$2x^3 + 11x + 14$$

Subtract the following polynomials.

$$(29) (4x^3 - 9x^2 - 2x + 3) - (6x^3 - 7x^2 + 2x - 10)$$

$$4x^3 - 9x^2 - 2x + 3 - 6x^3 + 7x^2 - 2x + 10$$

$$-2x^3 - 2x^2 - 4x + 13$$

$$(30) (10x^4 + 3x^3 - 7) - (8x^3 + 2x^2 - 3x + 2)$$

$$10x^4 + 3x^3 - 7 - 8x^3 - 2x^2 + 3x - 2$$

$$10x^4 - 5x^3 - 2x^2 + 3x - 9$$

$$(31) (-5x^3 - 2x^2 + 6x + 5) - (3x^4 - 6x^3 + 2x^2 - 3x - 3)$$

$$-5x^3 - 2x^2 + 6x + 5 - 3x^4 + 6x^3 - 2x^2 + 3x + 3$$

$$-3x^4 + x^3 - 4x^2 + 9x + 8$$

$$(32) (-4x^3 + x) - (-2x^3 + 6x - 9)$$

$$-4x^3 + x + 2x^3 - 6x + 9$$

$$-2x^3 - 5x + 9$$

Multiply the following polynomials.

$$(33) 8x^2(3x^4 - 2x + 5) = 24x^6 - 24x^3 + 40x^2$$

$$(34) -2z^3(2z^4 - 8z^2 + 3z^3 - 4z - 9) = -4z^7 + 16z^5 - 6z^6 + 8z^4 + 18z^3$$

$$(35) 3y^4(y^3 - 5y^2 - y + 7) = 3y^7 - 15y^6 - 3y^5 + 21y^4$$

$$(36) 3x^8(3x^4 - 2x^3 - 7) = 9x^{12} - 6x^{11} - 21x^8$$

$$(37) (x - 2)(x + 6) = x^2 + 6x - 2x - 12 = x^2 + 4x - 12$$

$$(38) (4x + 3)(3x + 7) = 12x^2 + 28x + 9x + 21 = 12x^2 + 37x + 21$$

$$(39) (x - 1)(x - 5) = x^2 - 5x - x + 5 = x^2 - 6x + 5$$

$$(10) (x+2)(x-8) = x^2 - 8x + 2x - 16 = \boxed{x^2 - 6x - 16}$$

$$(11) (x+3)(x^2+1x-1) = x^3 + 4x^2 - x + 3x^2 + 12x - 3 \\ = \boxed{x^3 + 7x^2 + 11x - 3}$$

$$(12) (x-5)(2x^2-x+6) = 2x^3 - x^2 + 6x - 10x^2 + 5x - 30 \\ = \boxed{2x^3 - 11x^2 + 11x - 30}$$

$$(13) (x+10)(x^2-7x+5) = x^3 - 7x^2 + 5x + 10x^2 - 70x + 50 \\ = \boxed{x^3 + 3x^2 - 65x + 50}$$

$$(14) (2x-1)(3x^2+8x+11) = 6x^3 + 16x^2 + 22x - 3x^2 - 8x - 11 \\ = \boxed{6x^3 + 13x^2 + 14x - 11}$$

Divide the following polynomials.

$$(15) \frac{10x^4 + 20x^5 - 16x^4 + 4x^3}{2x^4}$$

$$\boxed{5x^4 + 10x^3 - 8x^2 + 2x}$$

$$(16) \frac{21z^{10} + 9z^5}{3z^5}$$

$$\boxed{7z^5 + 3}$$

$$(17) \frac{-36y^7 - 27y^6 + 45y^5 - 81y^4}{-9y^3}$$

$$\boxed{4y^4 + 3y^3 - 5y^2 + 9y}$$

$$(18) \frac{25x^5 + 60x^4 - 35x^3 - 70x^2 + 40x}{-5x}$$

$$\boxed{-5x^4 - 12x^3 + 7x^2 + 14x - 8}$$

A Review of Arithmetic Long Division

Example I) $7,491 \div 3$

$$\begin{array}{r} 2497 \\ 3 \overline{) 7,491} \\ \underline{6} \\ 14 \\ \underline{12} \\ 29 \\ \underline{27} \\ 21 \\ \underline{21} \\ 0 \end{array} \quad \boxed{2,497}$$

Example II) $2,135 \div 8$

$$\begin{array}{r} 266 \\ 8 \overline{) 2,135} \\ \underline{16} \\ 53 \\ \underline{48} \\ 55 \\ \underline{48} \\ 7 \end{array} \quad \boxed{266 + \frac{7}{8}}$$

Example III) $5,576 \div 4$

$$\begin{array}{r} 1394 \\ 4 \overline{) 5,576} \\ \underline{4} \\ 15 \\ \underline{12} \\ 37 \\ \underline{36} \\ 16 \\ \underline{16} \\ 0 \end{array} \quad \boxed{1,394}$$

Example IV) $1,287 \div 5$

$$\begin{array}{r} 257 \\ 5 \overline{) 1,287} \\ \underline{10} \\ 28 \\ \underline{25} \\ 37 \\ \underline{35} \\ 2 \end{array} \quad \boxed{257 + \frac{2}{5}}$$